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DATA SCIENCE

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Independent Project Presentation

On

*LIBRARY MANAGEMENT
SYSTEM*



Content

- Introduction
- Problem Statement
- Objectives
- Literature Review
- Implementation & Design
- Results / Output
- Challenges
- Limitations
- Conclusion
- Future Work and Scope
- References



Introduction

A Library Management System (LMS) is a software application used to manage library activities such as adding books, issuing books, returning books, and maintaining student records. In a college library, handling all these tasks manually is difficult and time-consuming. Therefore, a computerized Library Management System using Python is developed to make library operations simple, fast, and accurate.



Problem Statement

In many college libraries, library operations such as maintaining book records, issuing and returning books, and keeping track of students are still handled manually or using outdated systems. These traditional methods require a lot of paperwork and human effort, which often leads to **errors, data loss, duplication of records, and delays** in searching and updating information.

As the number of students and books increases, it becomes difficult for librarians to efficiently manage library resources. Finding the availability of a book, tracking issued books, calculating due dates, and maintaining accurate records consume a lot of time. Manual systems also lack proper security and do not provide quick access to information.



Objectives

- To automate library operations
- To reduce manual work and errors
- To manage book records efficiently
- To maintain member information
- To provide a secure login system
- To simplify book issue and return process
- To calculate fines automatically
- To generate reports
- To provide a user-friendly graphical interface
- To enhance practical programming skills



Literature Review

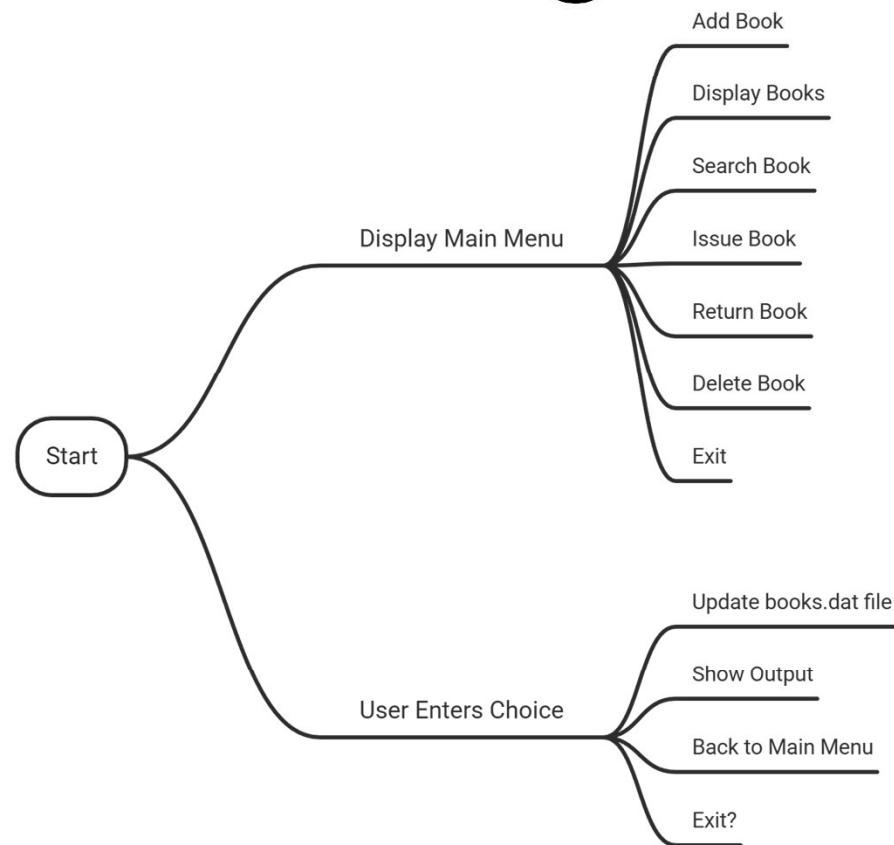
Early library systems were mostly manual, relying on registers and paper records to manage books and members. Studies show that manual systems are time-consuming, prone to errors, and inefficient when the size of the library increases. To overcome these limitations, computerized systems were introduced, initially using basic programming languages and file-based storage.

With the advancement of technology, database-driven library management systems became popular. Research indicates that using relational databases such as **SQLite** or **MySQL** helps in maintaining data consistency, reducing redundancy, and improving data retrieval speed. Many existing systems focus on automating core functions like book search, issue/return, and report generation.



Library Management System Flowchart

Implementation & Design





Results / Output

```
===== Library Management System =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice: 1

Enter Book ID: 1001
Enter Book Name: ADHURI
Enter Author Name: ANIL NAI
Enter Quantity: 5

Book Added Successfully!
```

```
===== Library Management System =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice: 3

Enter Book ID to search: 101

Book Found:

ID: 101
Name: RET
Author: ANIL NAI
Quantity: 10
-----
```

```
===== Library Management System =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice: 4

Enter Book ID to issue: 1001

Book Issued Successfully!
```

```
===== Library Management System =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice: 5

Enter Book ID to return: 1001

Book Returned Successfully!
```

```
===== Library Management System =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice: 6

Enter Book ID to delete: 1001

Book Deleted Successfully!
```



Challenges

- File Handling Complexity
- Maintaining Data Consistency
- Limited Error Handling
- Security Issues
- Scalability Problems
- Console-Based Interaction



Limitations

- No Database Integration
- Single-User System
- No Authentication Mechanism
- No Fine Management System
- No Date Tracking
- Risk of Data Corruption
- Limited Reporting Features



Conclusion

The Library Management System developed using Python provides an efficient and simple solution for managing basic library operations. This project successfully demonstrates how object-oriented programming, file handling using the pickle module, and menu-driven logic can be used to store and manage book records in a systematic manner.

The system allows users to add, display, search, issue, return, and delete books, which helps in reducing manual work and improving accuracy.



Future Work and Scope

- Database Integration
- Graphical User Interface (GUI)
- User Authentication System
- Fine Management System
- Date Tracking and History
- Multi-User Support
- Report Generation
- Backup and Recovery



References

- Books: Learning Python
Think Python
Head First Python
- Websites: YouTube



Thank you